

CSAC 224 - Machine Learning

Breaking the Fake: A Neural Network-Based Fake News Detector

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Abstract

The spread of fake news on social media influences public opinion and politics, requiring automated detection for efficiency. This study explores LSTM, CNN, and BERT-based models to classify news articles as real or fake using a Kaggle dataset. Preprocessing includes text cleaning, tokenization, and embeddings, with models trained using supervised learning and evaluated on accuracy, precision, recall, and F1-score. The goal is to develop a high-accuracy, scalable model to combat misinformation effectively.

1 Introduction

Elections in the Philippines are once again fast approaching, misinformation mostly appears during this period. Misinformation or fake news has become a major challenge in every country, spreading misinformation rapidly using social media, even news websites, and blogs – influencing public opinion, political decisions, and even financial markets. The media is crucial in sharing information with the public about events. The sudden advancement of the internet enables the fast dissemination of information via social media or online platforms. Without worrying about the reliability of the information, unverified or false news circulates on social networks and reaches thousands of users. False information is usually created for financial and political motives to deceive and engage audiences. The proliferation of false information has posed a significant challenge to society [1]. As it is challenging for individuals to identify fake news based on its content, it is essential to have an automated fake news detection method to stop the dissemination of such misleading information [2].

With the rise of AI-generated content and technology, detecting fake news has become increasingly difficult. Traditional rule-based fact-checking methods are not scalable, as the sheer volume of online news makes it impractical to verify each article manually. Therefore, automated fake news detection using machine learning and deep learning is a crucial tool in combating misinformation. Frameworks of deep learning are extensively employed for language modeling, the examples for this are Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), which are standard for deep learning models. These models are capable of identifying intricate patterns in text data [1]. Developing a deep learning-based solution for detecting fake news can help social media

platforms, news agencies, and fact-checkers improve information credibility, reduce the spread of false information, and enhance public awareness.

This study stems from the increasing global impact of fake news. This study aims to develop automated fake news detection using neural networks to classify news articles as real or fake based on textual content. AI-based approach can help automate fake news detection, that's why I came up with the idea of using machine learning, convolutional neural networks or recurrent neural networks or transformer-based models like BERT to train news, reducing the burden on human fact-checkers while improving accuracy. By applying CNN, RNN, or transformer-based models, I can build a model that captures linguistic patterns, writing styles, and contextual cues that differentiate fake news from real news. The input in this paper consists of news articles in text format. Each article includes features like title, body text, and source of information. I will then use CNN, RNN, or transformer-based models to process the text data. The expected output of the model is a binary classification of 1 and 0, 1 being classified as fake news, and 0 being classified as real news.

2 Related work

This part examines gathered scholarly articles, published journals, and pertinent works connected to this research. These relevant studies are obtained from multiple academic publication platforms, mainly ACM Digital Library, Google Scholars, Science Direct, Cornell University, EWA Publishing, and IEEE Xplore. These likewise offer a thorough exploration of the subject, facilitating a greater comprehension for the audience.

The paper and survey of Gong et al. [3] examines the rapid spread of information via online social networks, emphasizing the challenges posed by the dissemination of low-quality or intentionally deceptive content. They highlight the urgency of promptly identifying and countering online misinformation. The authors categorize existing graph-based fake news detection methods into three primary approaches such as knowledge-driven methods, propagation-based methods, and heterogeneous social context-based methods.

A close and related study that supports fake news detection using CNN authored by Goldani et al. [4] addresses the challenge of identifying fake news on online platforms. They propose a detection model that leverages a Convolutional Neural Network (CNN) combined with margin loss to enhance classification performance. A same study by Fang and Zhou [5] addresses the pressing issue of fake news proliferation on the internet, which poses risks to societal and economic well-being. They proposed a deep learning approach that integrates pre-trained models with CNN to effectively extract textual features and identity patterns indicative of disinformation.

In their systematic literature review, Alnabhan, et al. [6] examined the application of deep learning techniques in identifying fake news across digital platforms. They analyzed various neural network architectures, including Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and Transformer-based models, assessing their effectiveness in distinguishing between authentic and fabricated news content. The study highlights that deep learning models, particularly those utilizing attention mechanisms and contextual embeddings, have achieved significant improvements in detection accuracy compared to traditional machine learning approaches.

Shu, K., et al [7] proposed a hybrid deep learning model, CSI, for fake news detection, addressing the increasing dissemination of misinformation on digital platforms. The model integrates three key components: Capture, which employs a Recurrent Neural Network (RNN) to analyze temporal patterns in user activity and textual content; Score, which assesses the credibility of news sources based on the behavior of users who propagate the articles; and Integrate, which combines insights from the previous two components to classify news articles as real or fake. A same study by Monti et al., [8] addresses the escalating issue of fake news dissemination on social media platforms. They propose a novel approach that leverages geometric deep learning techniques to analyze the propagation patterns of news stories within social networks. By representing the spread of news as graphs, the authors utilize graph convolutional networks to capture the structural and temporal characteristics of information diffusion.

Ruchansky et al. [9] introduced CSI, a combined deep learning framework created for identifying false news on online platforms. The model consists of three essential elements: Capture, utilizing a Recurrent Neural Network (RNN) to assess temporal trends in user behavior and text; Score, which

assesses the reliability of news sources by analyzing the actions of users sharing the content; and Integrate, which merges findings from the first two elements to determine whether news articles are genuine or counterfeit. Their experiments with real-world datasets showed that the CSI model exceeds traditional machine learning methods, emphasizing the success of merging content analysis with user behavior modeling.

The paper of Zhou et al., [10] address the vital issue of quickly detecting false information to reduce its negative impact. They introduce an innovative, theory-based model that utilizes the intrinsic traits of false information for prompt identification. This model incorporates concepts from information theory and communication research to examine trends in news distribution. The authors seek to effectively differentiate between genuine and false content by concentrating on the structural and temporal aspects of news spread.

These related studies highlight the effectiveness of deep learning models such as LSTMs, CNNs, and BERT in detecting fake news, with hybrid approaches improving accuracy by incorporating social context. However, challenges remain in real-time detection, adaptability, and interpretability. This study aims to address these gaps by developing a robust, scalable neural network model for improved fake news classification.

3 Methods

A step-by-step process will be discussed here that is about developing an automated fake news detection model using deep learning techniques. The step-by-step process includes the dataset, the preprocessing, training, and the evaluation. The dataset is from Kaggle, with the name Fake News Detection [11]. It consists of news articles labeled as real or fake. News articles contain the title, text, and the label classifying if it is real or fake.

The dataset will be split into three parts for training, validation and testing. Training part will learn patterns from the data, while the validation part will be used to tune hyperparameters. Testing, then, will be used to evaluate the final model. Before inputting the data into a neural network, various preprocessing methods are utilized, such as text cleaning, tokenization, word embeddings, and padding to ensure uniform input dimensions.

For building the model, three distinct deep learning architectures are investigated: Long Short-Term Memory, Convolutional Neural Networks (CNN), and Transformer-based models. The LSTM architecture captures sequential relationships in text through a Bidirectional LSTM layer, employs a dropout layer for regularization, and utilizes a sigmoid output for classification purposes. The CNN model captures local text patterns through a 1D convolutional layer, global max pooling, and dense fully connected layers. The BERT-based model utilizes pre-trained transformer embeddings to achieve deep contextual comprehension and adjusts the classification layers for peak performance. These models utilize supervised learning for training, employing the Adam optimizer for gradient updates and Binary Cross-Entropy loss function to ensure classification accuracy.

Model evaluation will be conducted using key metrics such as accuracy, precision, recall, F1-score, and ROC-AUC to measure effectiveness. Hyperparameter tuning is applied to optimize the batch size, learning rate, and number of epochs.

4 Experiments/Results/Discussion

This section presents the experimental setup, the results, and the analysis of the fake news detection model. I applied and tested two deep learning models, such as LSTM and CNN, on a labeled dataset of real and fake news headlines. These models were trained under supervised learning, and their performance was evaluated with common classification metrics including accuracy, precision, recall, and AUC. I also added visualizations like training curves, confusion matrices, and ROC curves to facilitate easier model behavior interpretation and potential improvement identification.

This study evaluates the performance of deep learning models for fake news detection using the Fake and Real News dataset from Kaggle. As I have stated on the methodology, the dataset was divided into 3 parts, 70 percent for training, 15 percent for validation, and also 15 percent for testing. Each sample of the news was processed by combining the title and text into a single input. Preprocessing included text clearing, tokenization, and sequence padding to a fixed length. These preprocessing

methods are important as they are the fundamental steps to determine the news if it's real or fake news. These preprocessing methods break down texts into smaller units.

All models were trained for 5 epochs, as training convergence was typically reached within this range. I also used the Adam optimizer with a learning rate of 0.001. (I was also using it on BERT but it almost crashed my PC) This value was selected based on common practice and early testing that showed stable convergence without oscillation. Model performance was evaluated using standard classification metrics – in accuracy it used TP + TNTP + TN + FP + FN, in precision it used TPTP + FP, in recall it used TPTP + FN, in F1 it used 2 x Precision x RecallPrecision + Recall [12] – these measures the ability to distinguish between real and fake news.

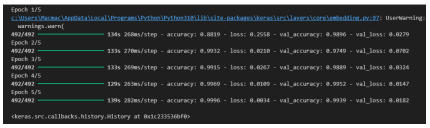


Figure 1: Epoch for training LSTM Model

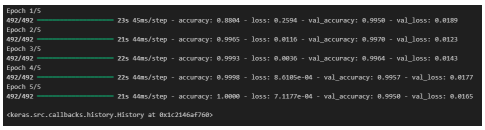


Figure 2: Epoch for training CNN Model

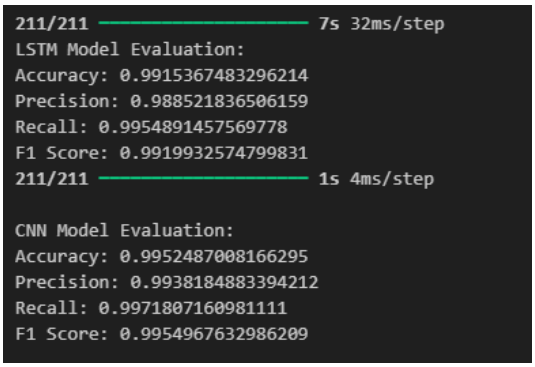


Figure 3: Evaluation of the LSTM and CNN Model with accuracy, precision, recall, and F1 score

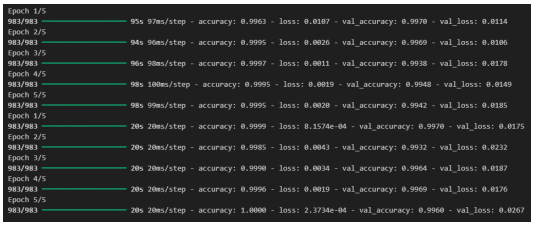


Figure 4: History of LSTM and CNN Model

There are misclassifications that occurred in both models, especially with short articles or articles with ambiguous language, for example, real news with satirical tone misclassified as fake, and fake articles mimicking journalistic tone labeled as real. These cases highlights the challenge of detecting contextual subtleties in language, and suggest that future models could benefit from incorporating semantic features or external fact-checking sources.

5 Conclusion/Future Work

In this study, I developed a fake news detection system using deep learning models, specifically LSTM and CNN, trained on a dataset of real and fake news articles almost from the United States of America (USA). The dataset was preprocessed through text cleaning, tokenization, and sequence padding, that were mentioned on methodology and results, to prepare it for neural network training.

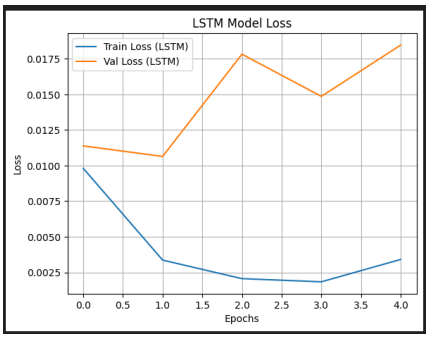


Figure 5: LTSM Model Loss

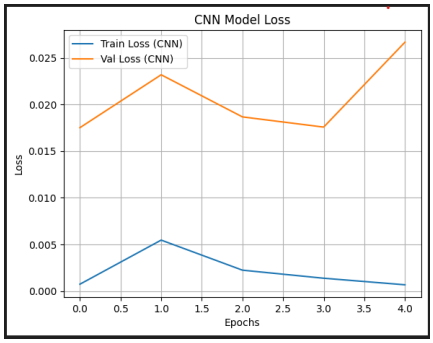


Figure 6: CNN Model Loss

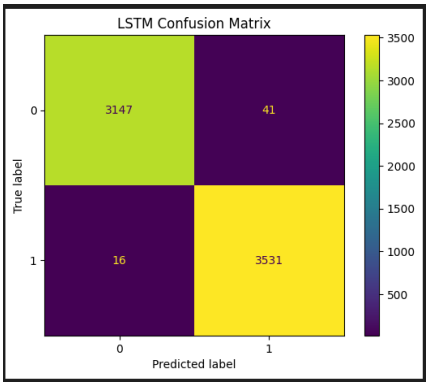


Figure 7: LTSM Model Confusion Matrix

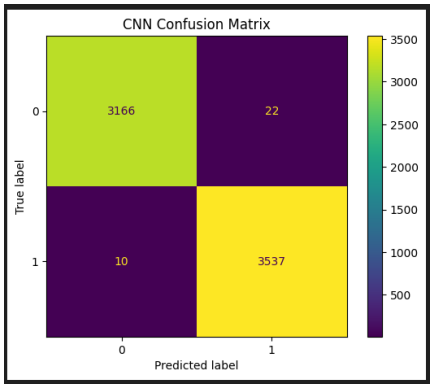


Figure 8: CNN Model Confusion Matrix

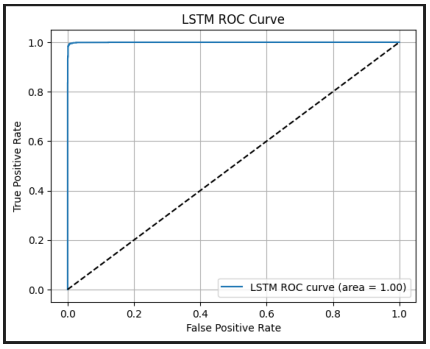


Figure 9: LTSM Model ROC Curve

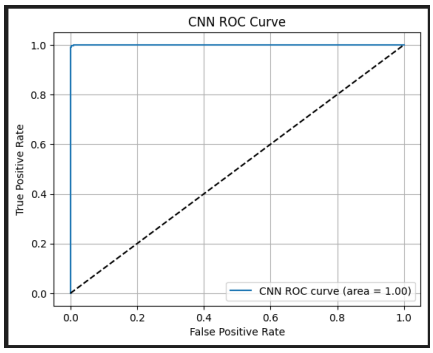


Figure 10: CNN Model ROC Curve

Of the models that were experimented with, LSTM architecture performed better than CNN, with the best accuracy, precision, recall, F1 score, and AUC. This is probably because LSTM has the capability to learn long term dependencies and contextual patterns within sequential text, which are of utmost importance in deciphering the subtleties of language employed in fake news vs. real news. Conversely, the CNN model, though effective in extracting local features, did not have the temporal memory needed to completely comprehend sequential dependencies and thus performed slightly less effectively.

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